SHAMKUMAR M

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CAREER OBJECTIVE

Motivated Embedded Systems Engineer with a solid foundation in C programming, Digital Electronics, and microcontroller-based systems. Proficient in debugging and testing embedded applications with hands-on experience in IoT-based system development. Familiar with integrating sensors, actuators, and communication modules for real-time monitoring and automation. Interested in applying Embedded Systems, IoT, and AI concepts to develop intelligent hardware and system-level solutions.

SKILLS

Programming Languages

- C Programming
- Embedded C
- Python

Electronics & Hardware

- Digital Electronics
- Microprocessors (8085 Basics)
- Microcontrollers (8051 /
- ESP32 Basics)
- Sensor Interfacing
- Analog & Digital Circuits
- Electronic Circuit Design

Embedded & IoT

- Basic IoT Prototyping
- Networking Basics (TCP/IP)

Tools

- Arduino IDE
- MATLAB
- VS Code
- Git (Basic)

AREA OF INTEREST

- Embedded Systems Programming
- Internet of Things
- Software Development

ACHIEVEMENTS

- Awarded 2nd Place in **Logo Design at ELECTROBLITZ** the **UNECS & IETE Students Forum**, for effectively representing the ECE department.
- **Secured 1st rank** in a competitive quiz event organized by **National Service Scheme (NSS)** at Sona College of Technology. Recognized by faculty and NSS program officers.

EDUCATION

- **BE-ELECTRONICS AND COMMUNICATION ENGINEERING [2023-2027]**
  SONA COLLEGE OF TECHNOLOGY - CGPA: 8.33/10
- **HSC [2021-2022]**
  MANGALAM HSS ANTHIYUR -83%
- **SSLC [2019-2020]**
  MANGALAM HSS ANTHIYUR -78.6%

INTERNSHIP

Titan Engineering & Automation Ltd. | Intern [May 2026 – June 2026]

Project: BOT-GRT – High-Speed Graphite Pick & Place Automation System

- Learned industrial automation and gantry motion-control systems.
- Observed machine-vision inspection for component placement accuracy.
- Gained exposure to barcode traceability and factory data integration.

PROJECTS

Smart Fish Farming (Aquaculture) Monitoring System

Tools Used: ESP32, Embedded C, Blynk IoT, DS18B20, Water Level Sensor

- Developed an IoT-based **smart aquaculture monitoring system** using **ESP32** to monitor **water temperature** and level in real time.
- **Automated water pump** control and **fish feeding based on sensor data.**
- **Integrated Blynk IoT** and **OLED display** for live monitoring and **improved farm efficiency.**

CERTIFICATIONS

- **NPTEL –** Industry 4.0 & IIoT (Elite + Gold, Topper)
- **C-Programming Foundation Course –** Etalvis Learning (July 2025)
- **Introduction to Internet of Things –** Cisco Networking Academy
- **MATLAB & Simulink Onramp -** MathWorks
- **Exploring C –** University of Michigan (Coursera)
- **Basics of Python –** Infosys Springboard
- **Foundations to Communication Mastery –** Udemy

CO-CURRICULAM ACTIVITIES

- **Presented Research** at International Conference on Sustainable Materials & Regenerative Tech (SMART-2025) in collaboration with IACG, Chennai.
- **Awarded a certificate** in the National Level Srinivasa Ramanujan Mathematical Competition (Levels 1 & 2) organized by ISTE Tamil Nadu Section (2024).
- **Selected at the college level for Smart India Hackathon (SIH)** 2025, working on an AI-based Automatic Backpack Pesticide Sprayer solution.
- **Certified in Advanced Technical Skills by IMAGECON Academy**(NASSCOM Member, 2024).
- **Selected as an Internshala Student Partner (ISP)**, contributing to student outreach by promoting internships and online training programs.